

A 500-MS/s 8-bit SAR ADC Generated from an Automated Layout Generation Framework in 14-nm FinFET Technology

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ABSTRACT

This paper presents the process of generating the layout of the 8-bit Successive Approximation Register Analog to Digital Converter (SAR ADC). By utilizing LAYGO2 [1], a Python framework that allows for detailed and flexible specification of custom layout generation processes, code-based generators for the component blocks of a SAR ADC were developed according to their specific operational characteristics and requirements. As a result, 85.5% of the SAR ADC was automatically generated. The SAR ADC test chip was fabricated in a 14-nm CMOS FinFET process and achieved an SNDR of 41.67 dB at 500 MS/s, consuming 2.07 mW from 0.9 V supply, and occupying an area of 4,131 um^2.

CCS CONCEPTS

• Hardware~Electronic Design Automation~Methodologies for EDA~Software tools for EDA

KEYWORDS

FinFET, LAYGO2, layout generation, SAR ADC

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1 Introduction

Due to the rapid growth in the field of artificial intelligence, demands on high-performance System-on-a-Chip (SoC) have increased significantly, driving competition to use cutting-edge processes and resulting in substantial human and temporal resources expended on chip design. SoC design companies aim to enhance chip performance using finer technologies for their chip fabrication. However, with advanced processes such as FinFET and gate-all-around FET (GAAFET), the complexity of design rules increases drastically, thereby prolonging the development cycle due to increased design time [2], [3]. While various EDA tool companies have continually enhanced design productivity in standard digital circuits through automation and AI technologies, the design of custom circuits such as Analog and Mixed-Signal (AMS) circuits, high-speed digital circuits, and memory circuits, which utilize full-custom design methodologies, presents increased difficulty. This escalates the time and cost involved in circuit fabrication. Specifically, Analog-to-Digital Converter (ADC) circuits that sample external analog signals at high rates and convert them into digital signals are particularly affected by these challenges. They must be integrated into the same advanced processes as their digital counterparts to utilize their output digital signals. Despite the use of Capacitive Digital-to-Analog Converter (CDAC)-based Successive Approximation Register (SAR) architecture, which is suitable for digital processes, fundamental issues related to custom design methodologies remain unresolved.

This paper introduces a method to address the productivity issues in AMS circuit design in advanced processes, by generating the layout of a representative analog circuit, the SAR ADC, using the generator-based layout automation framework, LAYGO2 [1]. It also discusses the features of LAYGO2 that are specialized for analog layout generation. Moreover, the paper includes the measurement results of the SAR ADC fabricated in a 14-nm FinFET process, verifying that the proposed method can generate a practical ADC layout that achieves a 500 MS/s sampling rate.

2 Structure of Generated SAR ADC

SAR ADCs are frequently used to achieve high sampling rates in ultra-high-speed Serializer-Deserializer (SerDes) applications due to their high conversion speeds and resolutions, and their digital-friendly structures, often in conjunction with the time interleaving technique [4], [5]. However, similar to other ADC circuits, their analog components are sensitive to layout-dependent effects, requiring careful attention not only in schematic design but also during layout generation. The basic structure of the SAR ADC presented in this paper is illustrated in Fig. 1. The ADC primarily



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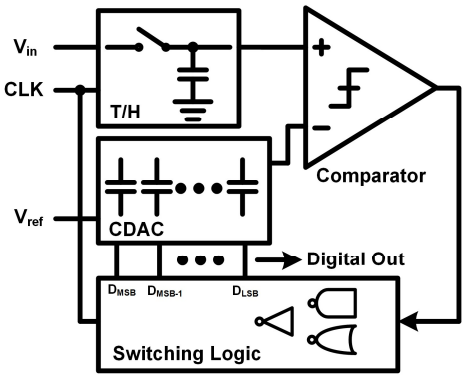


Figure 1: SAR ADC basic structure.

consists of a differential CDAC, comparator, track-and-hold, and switching logic. The ADC operates by sampling the input signal through the track-and-hold circuit during each sampling cycle, performing bit-by-bit comparison operations with the comparator, and generating the reference voltage for the next comparison through the switching logic and CDAC, thus executing successive approximation. Below, we describe the key performance indicators and design considerations for each component affecting the overall performance of the ADC.

CDAC: The uniformity of the CDAC's capacitances is a critical factor that determines the ADC's linearity and harmonic characteristics. Major aspects influencing the CDAC non-uniformity include parasitic elements and layout mismatches. Layouts must be designed to minimize these parasitic and mismatch effects, adopting repetitive and symmetric patterns.

Comparator: The comparator compares the voltage output from the CDAC with a reference voltage for approximation. Typically composed of a sensing node followed by a regeneration stage, it is designed to have a high gain for sensitive comparison operations and is very susceptible to systematic offsets caused by layout mismatches. Therefore, it must be fabricated in a structure that minimizes the differential mismatch effects.

Track-and-hold: The track-and-hold (T/H) circuit is responsible for sampling the input analog signal and maintaining the sampled voltage while the subsequent SAR circuitry operates. Structurally, bootstrap switches, which show better linearity and harmonic characteristics than typical MOS switches or transmission gates, are commonly used [6].

Switching Logic: This switching logic receives outputs from the comparator and controls the CDAC for binary searching. Composed of digital logic gates, it often has a repetitive structure that increases with the number of bits, offering a high potential for productivity enhancement through automation.

3 Layout Generation of SAR ADC

Commonly, manually constructed layouts are known to have better

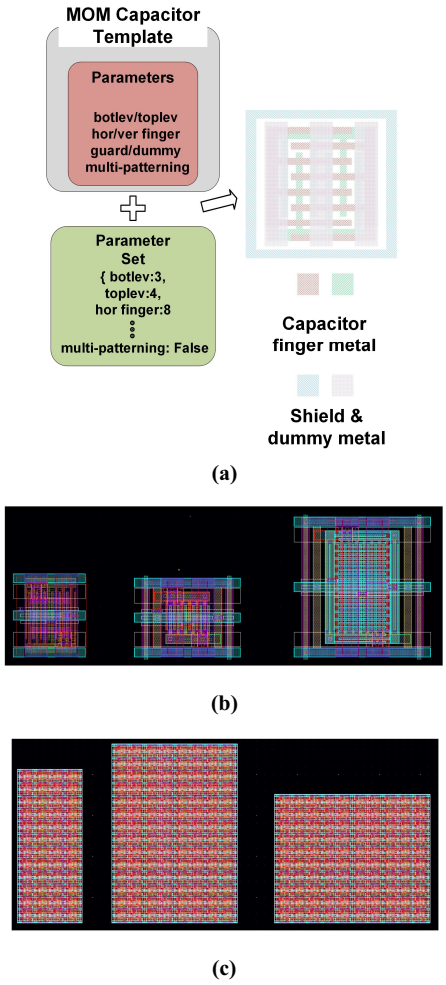
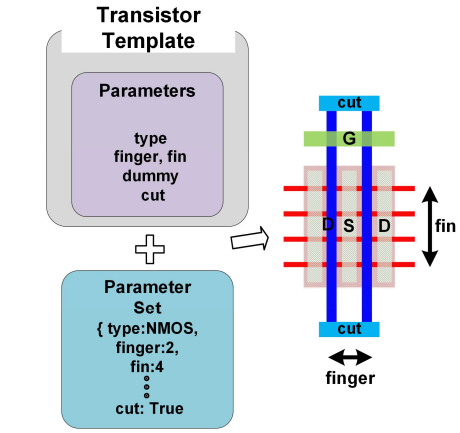
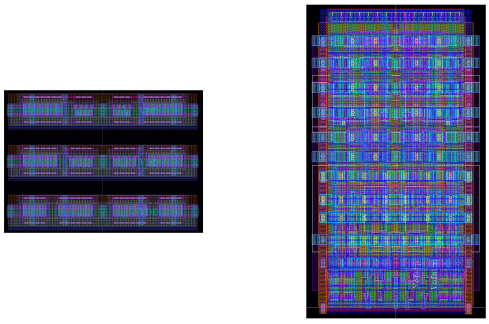


Figure 2: (a) The process of generating unit capacitor layouts, (b) examples of generated unit capacitors with various options, and (c) examples of generated CDAC.

matching and symmetry characteristics than automatically placed and routed layouts by electronic design automation (EDA) tools, making them more suitable for Analog and Mixed-Signal (AMS) circuits. However, manual layouts are also prone to errors during placement and metal routing due to the limitations of human effort, which can deteriorate their electrical characteristics. In contrast, utilizing code-based generators that describe detailed placement and routing pattern topologies can significantly reduce the occurrence of systematic mismatches and offsets. For instance, in symmetry-critical components such as differential pairs, using the same pattern code statements can ensure repetitive patterns and high symmetry, potentially reducing offsets caused by device placement and metal routing differences. Moreover, routing and placement can be efficiently conducted with parametrization, enhancing design reuse and productivity. Based on these



(a)



(b) (c)

Figure 3: (a) The process of generating transistor layouts, (b) examples of generated sub-cell of comparator with various options, and (c) example of generated comparator layout.

advantages, the code-based layout generation method for the SAR ADC in a FinFET process is described block by block in the following paragraphs.

First of all, the CDAC layout was generated by regularly placing unit capacitors, following the pattern guided by code. The characteristics of the CDAC significantly influence the performance of the SAR ADC [7]. Variations during operation or manufacturing processes can increase mismatch and nonlinearity. To minimize these nonideal effects, it is recommended to use a common centroid structure for placement. Additionally, appropriate use of dummy and shielding metals can minimize the mismatch and gradient effects. While unit capacitors often use parametrized cells (*p-cells*) provided by the Process-Development Kit (PDK) provided by the target foundry, manually crafted custom capacitors are also frequently used to optimize CDAC switching time and mismatch characteristics. LAYGO2 has the capability to

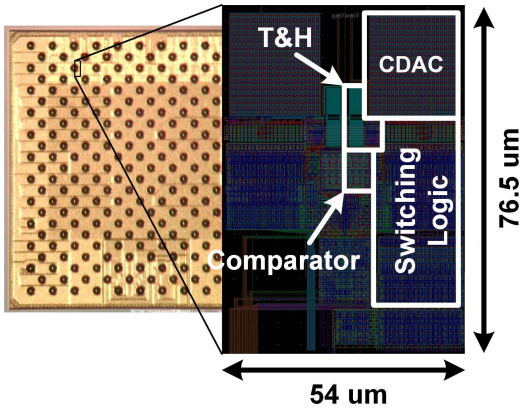


Figure 4: Die micrograph and layout of the generated SAR ADC.

create custom templates, *UserDefinedTemplates*, allowing for the generation of customized Metal-Oxide-Metal (MOM) capacitor templates. This feature was utilized to create compact unit capacitors, and LAYGO2's grid-based placement was employed to arrange the capacitor array. The process of CDAC layout generation with LAYGO2 is illustrated in Fig. 2(a). A custom MOM capacitor template was created with adjustable options for metal levels, number of fingers (capacitor size), guard/dummy metal, and multi-patterning. By inputting the desired parameters into this template, various forms of unit capacitors can be generated. Utilizing this information, CDACs with different aspect ratios can be created, as demonstrated in Fig. 2(b) and Fig. 2(c).

For the comparator, a Strong-Arm latch topology was used for its high-speed operation. During the sampling phase, the Strong-Arm latch compares the differential signal at the input pair and amplifies the sampled voltage difference during the regeneration phase [8]. The sizing optimizations regarding clock-to-q delay, noise, offset, power consumption, and layout area are best assessed after completing the layout. Sweeping each transistor sizing parameter during layout generation for comparative analysis is performed to optimize the characteristics. The layout placement is done such that the differential devices are placed symmetrically to reduce the systematic offsets. As previously mentioned, the Strong-Arm latch comparator is a sensitive circuit that requires a high degree of layout flexibility. Therefore, it is necessary to create a transistor template that includes transistor sizing, placement, routing, and dummy options. However, in advanced processes, the structure of transistors is complex, involving not only metal, and diffusion layers but also cut layers and CAD-specific layers. Creating a template that all these requirements is challenging. To address this, we utilized the transistor *p-cells* provided by the PDK in our template. These provided *p-cells* serve as a foundational basis for the transistor structure while adhering to design rules. We then added the desired options to form the transistor template. The process of generating transistors using this template is illustrated in Fig. 3. For the track-and-hold circuit, a bootstrap switch structure

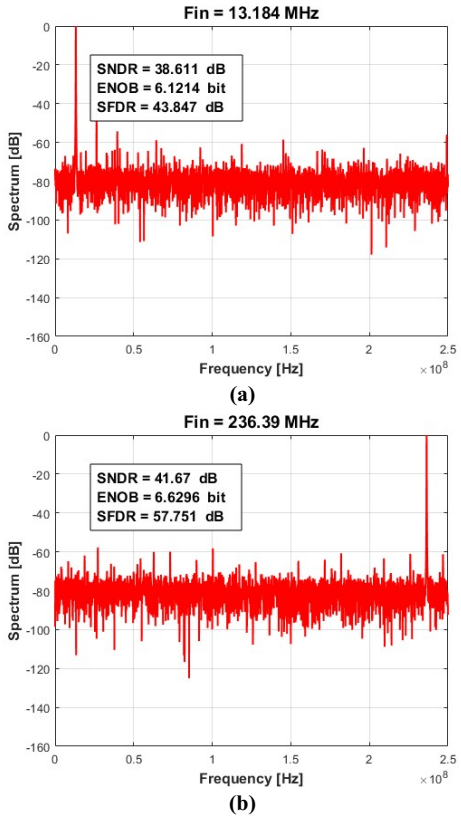


Figure 5: ADC measurement results: (a) Spectrum with low frequency and (b) near-Nyquist frequency inputs at 500MS/s.

was utilized to minimize the change in switch resistance values depending on the input signal. The arrangement of transistors and capacitors in the bootstrap switch needs to be fine-tuned to extremely optimize the performance of the switch, as it limits the maximally achievable performance of the ADC. Therefore, instead of completely automating the process of track-and-hold layout, manually crafted layouts were encapsulated into templates using LAYGO2's instance wrapping feature. These templates include information about the layout's size and pin configurations. This process facilitated merging with other blocks, enhancing placement and routing compatibility, while still yielding the opportunity to extremely optimize the layout structure.

Finally, the switching logic circuit is composed of multiple logic gates, which are regularly repeated, making it well-suited to layout generation via LAYGO2. The layout of the component logic gates was either generated by LAYGO2 from transistor level or handcrafted and wrapped by LAYGO2 for high-level generation, similar to the track-and-hold circuit, to extremely minimize area consumption. The placement of component gates and routing between gates are performed by LAYGO2, to utilize its design rule abstraction capabilities.

4 Measurement Results

The generated SAR ADC was fabricated in a 14-nm CMOS FinFET process. The die micrograph and layout of the SAR ADC are shown in Fig. 4. The total area of the SAR ADC is 4,131 um^2, and 85.5% of overall components are generated automatically. The manually designed layout includes deserializers and retimers, which can be readily automated in future versions. The SAR ADC achieves a sampling rate of 500 MS/s at 0.9-V supply voltage. The spectrum of ADC output for 1.5-V_{dpp} low-frequency and high-frequency sinusoidal inputs are depicted in Fig. 5. The ADC consumes 2.07 mW at 500 MS/s, which corresponds to 41.8-fJ/conv-step FoMW and 173.2-dB FoMS.

5 Conclusion

We demonstrated the operation of the SAR ADC fabricated in 14-nm FinFET, of which the layout was generated with LAYGO2. The LAYGO2's grid-based layout generation and flexible description capabilities are widely utilized to enhance productivity in advanced technologies. The generated ADC was fabricated to demonstrate the proposed methodology and achieved performance comparable to that of hand-crafted designs.

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